TIDBD-Lite Plasticity

Status: independent supporting mechanism study; not yet a positive performance proposal.

Abstract

This proposal studies per-feature step-size adaptation as a lightweight plasticity mechanism for streaming TD prediction. In a nonstationary sensor stream, feature relevance changes after a phase switch. A TIDBD-style learner should increase step sizes for newly useful features while keeping distractors quiet. The current main pilot shows this mechanism qualitatively: new-feature step size rises after the switch and distractor step size stays low. However, normalized TD still has lower late prediction error than the current TIDBD-lite implementation. The proposal is therefore a mechanism study, not a performance win.

Standalone Study Summary

This study examines per-feature step-size plasticity in a streaming prediction setting. The RL problem is not a full control task; it is a diagnostic stream where feature relevance changes and a learner must adapt without replay. The implemented method is TIDBD-lite, compared with fixed or normalized TD baselines. The experiment records prediction/TD error, feature-group behavior, per-feature alpha trajectories, and post-change adaptation. The current evidence shows visible step-size adaptation but no clear prediction-error advantage over normalized TD. The study is therefore a supporting mechanism diagnostic, not a positive performance proposal.

Research Motivation

Continual agents face changing feature relevance. A fixed global step size is a compromise: large enough to adapt quickly, but small enough not to destabilize irrelevant or noisy features. Per-feature step-size adaptation offers a more local form of plasticity.

The Alberta Plan treats step-size adaptation and feature utility as early building blocks for long-lived agents. This proposal asks whether a simple per-feature adaptive TD learner can detect changing relevance in the stream itself.

Research Question

Can per-feature step-size adaptation track changing feature relevance in streaming TD?

Hypothesis:

> After a relevance switch, a TIDBD-style learner should increase step sizes for newly relevant features, maintain lower step sizes for distractors, and recover prediction accuracy faster than fixed-alpha TD.

The current result supports the first two mechanism claims but not the stronger prediction error claim.

Alberta Plan Connection

The proposal connects to:

- - meta-learning of step sizes;
- - online prediction;
- - continual adaptation;
- - feature relevance tracking;
- - limited computation with linear function approximation.

It is not a deep plasticity study. The point is to inspect feature-wise learning-rate dynamics in a small streaming setting.

Related Work

TIDBD extends incremental delta-bar-delta ideas to TD learning with feature-wise step sizes. The Alberta Plan mentions per-weight step-size adaptation as part of early continual learning. Plasticity work in continual learning motivates recovery and feature-level diagnostics, though much of that literature uses deep networks outside this project.

Local references:

- - resources/alberta_plan_related/tidbd_1804.03334.pdf
- - resources/alberta_plan_related/alberta_plan_2208.11173.pdf

Environment

The setting is a nonstationary sensor prediction stream:

- - one feature group is relevant before a switch;
- - a different group becomes relevant after the switch;
- - distractor features remain mostly irrelevant;
- - the learner updates online without replay.

This is a mechanism environment. It creates controlled feature-relevance changes that can be measured directly.

Methods

Compared methods:

- - fixed TD with alphas 0.01, 0.03, 0.1;
- - normalized TD;
- - TIDBD-lite.

The local implementation is deliberately labeled TIDBD-lite. It is not claimed to be a full canonical reproduction of all TIDBD details.

Experimental Design

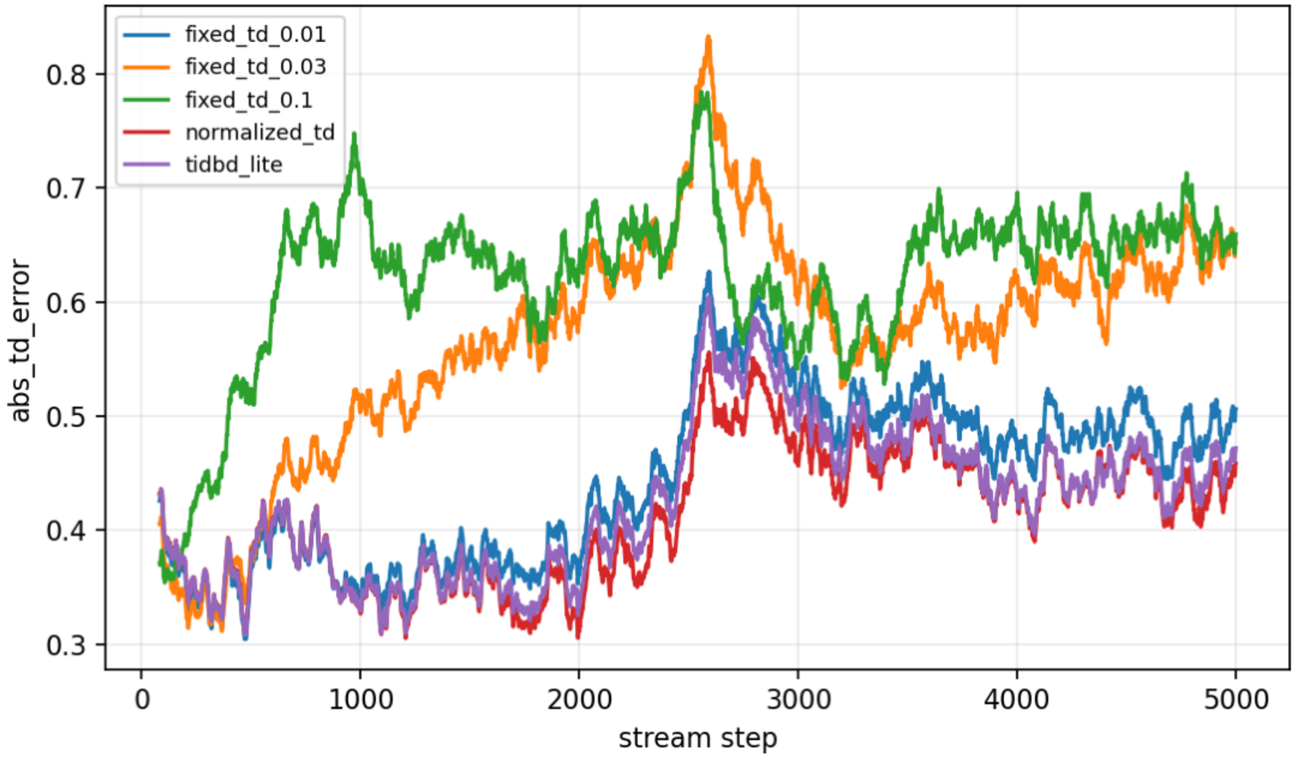
Current main pilot:

- - Seeds: 0-4.
- - Steps: 5000.
- - Switch: halfway through the stream.
- - Result path: experiments/alberta_core_rl/results/tidbd_plasticity/20260708T154941Z_main.

Metrics:

- - absolute TD error;
- - old-relevant feature step sizes;
- - new-relevant feature step sizes;
- - distractor step sizes;
- - recovery windows.

Primary figure:



Absolute TD error for TIDBD-lite and TD baselines.

Results

TIDBD-lite shows the intended feature-wise plasticity signal. New-feature step size rises from about 0.0068 before the switch to about 0.0093 in the late post-change window. Distractor step size remains near 0.0068.

However, normalized TD remains the stronger prediction-error baseline. Late post-change absolute TD error is about 0.4419 for normalized TD and about 0.4493 for TIDBD-lite. This means the mechanism signal does not yet justify a performance claim.

Analysis

The proposal separates two claims that are often conflated:

- - The learner changes step sizes in a sensible feature-specific direction.
- - That adaptation improves the prediction objective.

The first claim is supported. The second is not yet supported. This distinction makes the proposal scientifically useful: it identifies a mechanism that behaves plausibly but is not strong enough under the current environment and implementation.

Threats To Validity

The implementation is TIDBD-lite, not a canonical TIDBD reproduction.

The environment has one switch. Repeated changes may better expose plasticity advantages.

Normalized TD is a strong baseline because it controls update size directly; a fair final study should compare per-feature adaptation and output normalization more systematically.

The current metrics do not include feature correlation or utility contribution beyond step-size groups.

Reviewer Critique And Revisions

Plasticity reviewer:

• - Mean step size alone is insufficient.

Revision made:

• - Added group-wise old/new/distractor step-size logs and recovery windows.

Strict reviewer concern:

• - Do not call a local simplified algorithm "TIDBD" without qualification.

Revision made:

• - The report uses TIDBD-lite terminology.

Required next revision:

• - Implement canonical TIDBD or AutoStep, add repeated relevance switches, and report recovery AUC rather than only late error.

Conclusion

TIDBD-Lite Plasticity is a valid independent mechanism proposal, but not yet a positive performance story. It shows meaningful per-feature step-size adaptation after a relevance switch, while normalized TD remains stronger on prediction error. The next version should upgrade the algorithm and environment before making broader claims about plasticity.

Reproduction

```
cd /mnt/shared-storage-user/yupeng/Core-RL

PYTHONNOUSERSITE=1 MPLCONFIGDIR=/mnt/shared-storage-user/yupeng/Core-RL/.mplconfig \
  /data/yupeng/conda_envs/core-rl/bin/python experiments/alberta_core_rl/scripts/run_experiment.py \
  --config experiments/alberta_core_rl/configs/tidbd_plasticity/config_main.json
```